The iterated prisoner's dilemma: when does cooperation emerge?

Computational Intelligence (MAI): evolutionary computation practical work, exercise 8

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Abstract

We study whether cooperation emerges and persists in a population of memory-1 strategies for the iterated prisoner's dilemma evolved with a genetic algorithm, and how the answer depends on the level of action noise, $p_{\text{noise}} \in \{0, 0.01, 0.05\}$. With 30 independent runs per level, the final mean payoff drops from 2.98 (almost full cooperation, dominated by TFT) to 2.01 with $p_{\text{noise}} = 0.01$ and to 1.17 with $p_{\text{noise}} = 0.05$ ($\hat{A}_{12} = 1$ in all three comparisons). The initial hypothesis holds without noise and with high noise, but fails with moderate noise, where GRIM wins instead of the forgiving strategy PAVLOV. We explain this with the matrix of expected payoffs: with noise, PAVLOV can be invaded by ALLD and by GRIM, whereas GRIM and ALLD cannot be invaded by any memory-1 strategy. In a finite population with mutation, however, almost neutral paths connect them, and these produce cycles of cooperation and exploitation with $p_{\text{noise}} = 0.01$ and alternations between ALLD and GRIM with $p_{\text{noise}} = 0.05$.
As an extension, we repeat the experiment with memory-2 strategies (18 bits). With them, cooperation survives the noise: the median final payoff is 2.84 with $p_{\text{noise}} = 0.01$ and 2.55 with $p_{\text{noise}} = 0.05$ ($\hat{A}_{12} = 1$ against memory 1), because the most frequent final strategies are PAVLOV-like strategies that punish one extra round, and no memory-2 mutant can invade the most frequent one.

1 Question and hypothesis

In the prisoner's dilemma two players simultaneously choose to cooperate (C) or defect (D), with payoffs $T = 5$ (defecting against a cooperator), $R = 3$ (mutual cooperation), $P = 1$ (mutual defection) and $S = 0$ (cooperating with a defector). Since $T > R > P > S$, defection dominates in a single round; since $2R > T+S$, sustained cooperation is better than alternating exploitation. In the iterated version the same players play $N_r = 150$ rounds and can condition their action on the past, which makes reciprocity possible [2]. With *noise*, each intended action is flipped with probability p_{noise}; it is known that TIT-FOR-TAT (TFT) then gets trapped in mutual retaliation, whereas *win–stay, lose–shift* (PAVLOV) corrects it [6, 8].

Question. Does cooperation emerge and remain stable in an evolving population, and how does the answer depend on the noise level?

Hypothesis (stated before the experiments). We adopt the hypothesis proposed in the exercise, split into three falsifiable predictions:

H1 Without noise the population evolves towards TFT-like strategies and cooperation is almost full. *It would be refuted* if the final mean payoff were clearly below $R = 3$ (e.g. < 2.5) or if TFT were not the majority strategy.

H2 With moderate noise ($p_{\text{noise}} = 0.01$) forgiving, PAVLOV-like strategies dominate. *It would be refuted* if PAVLOV were not the most frequent cooperative strategy at the end of the runs.

H3 With high noise ($p_{\text{noise}} = 0.05$) cooperation collapses. *It would be refuted* if the final mean payoff did not approach $P = 1$ and defecting strategies did not dominate.

2 Experimental design

Representation. A memory-1 strategy is a 5-bit lookup table (0 = C, 1 = D): the first action and the response to each outcome of the previous round, CC, CD, DC, DD (own action first). There are only $2^5 = 32$

strategies, so they can be named and their frequencies tracked: ALLC `00000`, ALLD `11111`, TFT `00101`, PAVLOV `00110`, GRIM `00111` and STFT `10101` (TFT that starts by defecting). The rest are grouped as OTHER.

Fitness: exact expected payoff. Each individual plays against all the others (round robin, without playing against itself) and its fitness is the mean payoff per round, so that 3 corresponds to full cooperation and 1 to full defection. Since fitness depends on the rest of the population, this is coevolution, and the whole population is re-evaluated in every generation. With memory-1 strategies, the outcome of a round is a 4-state Markov chain (CC, CD, DC, DD) whose transitions depend on both strategies and on p_{noise}. We propagate the probability distribution over the four states for 150 rounds, which gives the exact *expected* payoff of each pair. This payoff equals the mean over infinitely many independent matches of 150 rounds, whereas our first implementation used the result of a single match played with random noise.

We thus compute, once per noise level, the 32×32 payoff matrix. This is preferable to simulating one noisy match per pair: a single match has a large variance (e.g. GRIM against GRIM with $p_{\text{noise}} = 0.01$: mean 1.66, standard deviation 0.55 across matches), and if it is reused for every copy of the same pair, a single stroke of luck decides the fitness of a whole class of strategies. Using 30 runs with different seeds would average part of this error out, but the exact expectation removes it from the fitness altogether.

Genetic algorithm and fixed parameters. We use DEAP [4] with the values suggested in the exercise (Table 1), which correspond to a standard GA: uniform crossover is natural because the 5 bits have no relevant order, and $p_m = 0.01$ per bit produces a mutant in about 5 % of the offspring of each generation, enough for the population to be continuously exposed to invading strategies. Elitism of 2 keeps the best strategy from being lost by chance.

Table 1: Parameters of the experiment. Only p_{noise} is manipulated.

Population	100	Selection	tournament of size 3
Generations	500	Crossover	uniform, $p_c = 0.7$
Rounds per match N_r	150	Mutation	bit-flip, $p_m = 0.01$ per bit
Payoffs (T, R, P, S)	$(5, 3, 1, 0)$	Elitism	2 individuals
Factor	$p_{\text{noise}} \in \{0, 0.01, 0.05\}$	Runs	30 per level (independent seeds)

Factor, control and response variable. The only factor is p_{noise}. The level $p_{\text{noise}} = 0$ is the *control*: its outcome is predictable from classical theory [2] (reciprocal strategies cooperate with each other and cannot be exploited). The response variable of each run is the mean payoff of the population averaged over the last 100 generations; averaging over time prevents the result from depending on the point of an oscillation at which the run ends. As a measure of the mechanism we record, in every generation, the frequency of the 32 strategies, and we call a *regime* a period of at least 10 generations in which the same strategy exceeds 50 % of the population.

Number of runs and statistics. A pilot of 10 runs per level gave standard deviations of 0.002, 0.08 and 0.21 for $p_{\text{noise}} = 0$, 0.01 and 0.05, with a 95 % confidence interval of the median of ± 0.17 in the most variable case. Although the differences between levels were already clear ($\hat{A}_{12} = 1$), with $p_{\text{noise}} = 0.05$ the distribution turned out to be bimodal (Section 3), so we set 30 runs per level, which reduces that interval to ± 0.08 and allows us to estimate the proportion of each outcome. Each level k uses seeds $100k, \ldots, 100k + 29$, so the samples are independent. For each pair of levels we report the difference of medians with a 95 % percentile bootstrap interval (10 000 resamples) [3], the Vargha–Delaney $\hat{A}_{12}$ statistic [7, 1] and the two-sided Mann–Whitney test with Holm's correction for multiple comparisons [5].

3 Results

Figure 1 shows the evolution of the mean payoff. Without noise, the median reaches almost full cooperation (> 2.95) in 9 generations and keeps it. With $p_{\text{noise}} = 0.01$ the median across runs oscillates around 1.7 with frequent peaks of cooperation (these peaks are why the per-run time average, our response variable, has a higher median, 2.01), and with $p_{\text{noise}} = 0.05$ it quickly drops to about 1.2. Table 2 and Figure 2 summarise the response variable: the three levels are completely separable ($\hat{A}_{12} = 1$ in all comparisons, i.e. any run with less noise beats any run with more noise), and the differences of medians are large compared with their

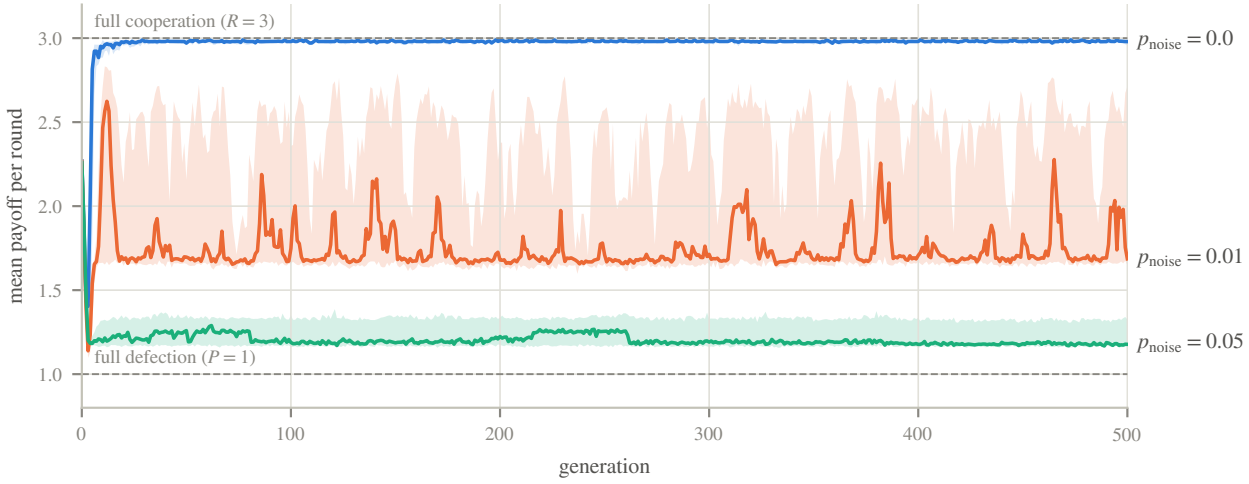


Figure 1: Mean payoff per round of the population over the generations: median (line) and interquartile range (band) of the 30 runs of each noise level.

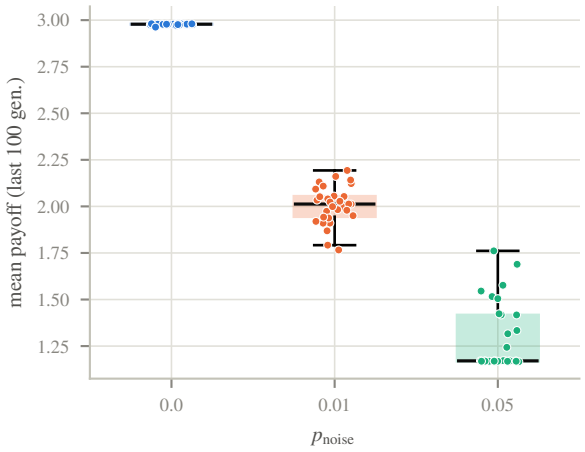


Figure 2: Response variable (mean payoff over the last 100 generations) of each of the 90 runs.

Table 2: Mean payoff over the last 100 generations (30 runs per level): median with 95 % bootstrap CI, and pairwise comparisons (Mann–Whitney p with Holm's correction).

p_{noise}	Median [95 % CI]	Min.–max.
0	2.978 [2.977, 2.979]	2.963–2.983
0.01	2.013 [1.976, 2.046]	1.766–2.193
0.05	1.171 [1.169, 1.325]	1.166–1.761
Comparison	**Diff. of medians [95 % CI]**	$\hat{A}_{12}$
0 vs 0.01	+0.966 [+0.932, +1.002]	1.00
0 vs 0.05	+1.808 [+1.653, +1.809]	1.00
0.01 vs 0.05	+0.842 [+0.679, +0.875]	1.00

In all three comparisons, $p_{\text{Holm}} = 9 \cdot 10^{-11}$.

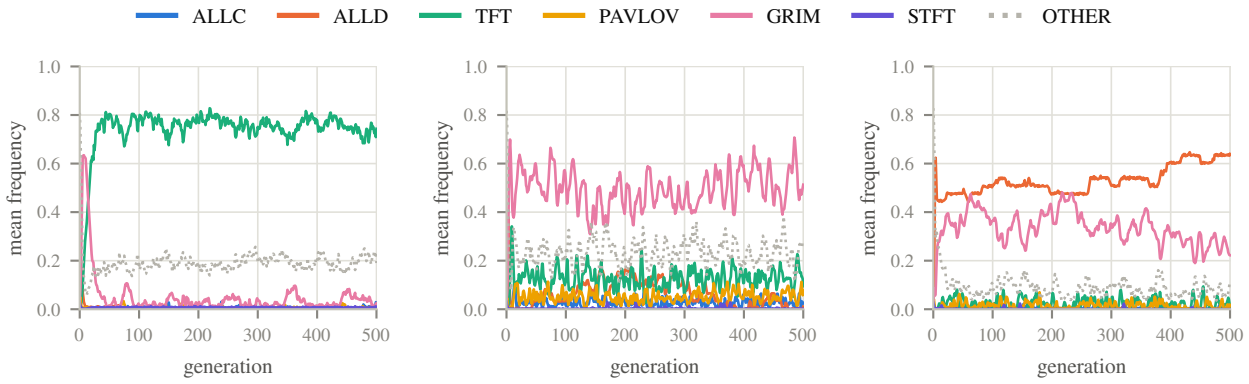
intervals. The intervals with $p_{\text{noise}} = 0.05$ are asymmetric because the distribution is bimodal: 18 runs are concentrated at 1.17 and the other 12 are spread between 1.24 and 1.76.

Which strategies win. Figures 3 and 4 show the composition of the population. We say that a strategy dominates a run if it exceeds 50 % of the population on average over the last 100 generations.

- $p_{\text{noise}} = 0$**:** TFT dominates in 29 of 30 runs (median 78 %) and exceeds 50 % of the population in 90 % of the generations from 100 onwards. Almost all the rest (OTHER, 17 %) is the genome `00100`, which differs from TFT only in its response to DD; among cooperators that entry is never used, so it is a neutral variant that drifts freely. **H1 is confirmed.**
- $p_{\text{noise}} = 0.01$**:** GRIM dominates in 19 of 30 runs (median 53 %), followed by TFT (13 %). PAVLOV does not exceed a median of 5 % and dominates no run. **H2 is refuted.**
- $p_{\text{noise}} = 0.05$**:** ALLD dominates in 19 runs (with 95 % of the population) and GRIM in 9; in the remaining 2 both coexist. The median payoff drops to 1.17. **H3 is confirmed**, although not every run ends in the same state.

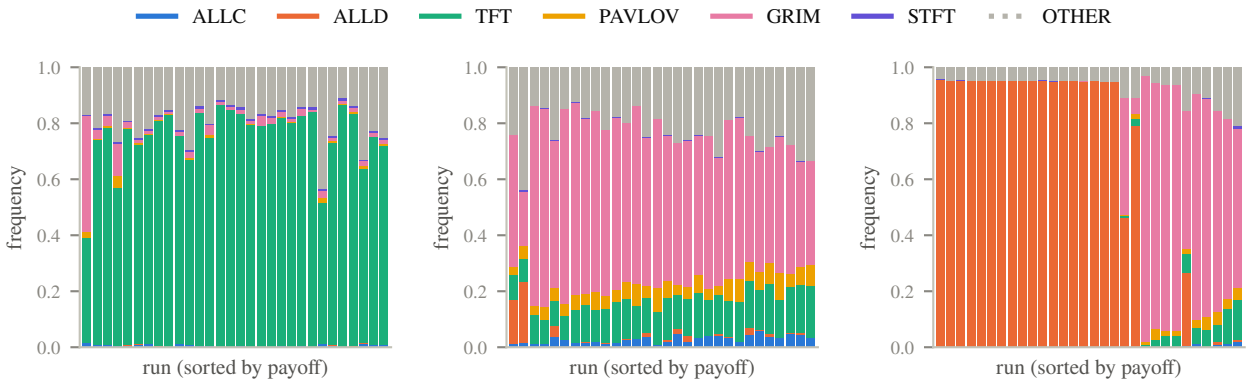
4 Mechanism

To explain these results we use the matrix of expected payoffs of the classical strategies (Figure 5), computed with the same exact model as the fitness. Altough there are results about non-named strategies mentioned in the following subsections that don't appear in the figure, they have been extracted from the complete results



(a) $p_{\text{noise}} = 0$ **(b)** $p_{\text{noise}} = 0.01$ **(c)** $p_{\text{noise}} = 0.05$

Figure 3: Frequency of the named strategies over the generations (mean of the 30 runs).



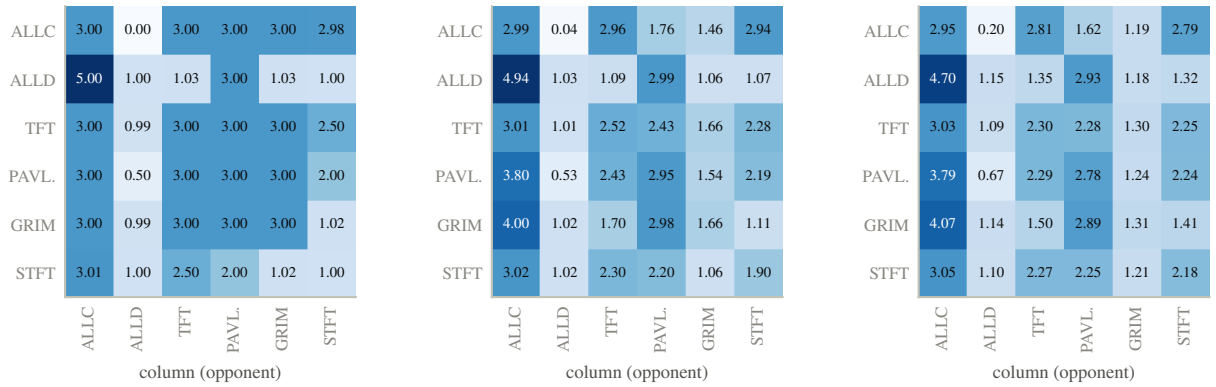
(a) $p_{\text{noise}} = 0$ **(b)** $p_{\text{noise}} = 0.01$ **(c)** $p_{\text{noise}} = 0.05$

Figure 4: Composition of each run, averaged over the last 100 generations. Each bar is a run, sorted from lowest to highest final payoff.

present in the project. In a large population made almost entirely of a resident strategy X, a mutant Y can invade it if $\pi(Y, X) > \pi(X, X)$, or if $\pi(Y, X) = \pi(X, X)$ and $\pi(Y, Y) > \pi(X, Y)$, where $\pi(A, B)$ is the payoff of A against B. Table 3 applies this criterion to each noise level. Moreover, since mutation usually changes a single bit, we also look at the advantage $\pi(Y, X) - \pi(X, X)$ of the 5 one-bit mutants of each resident: if it is almost zero, the mutant can spread by drift even if it does not invade.

Without noise. TFT, PAVLOV, GRIM and ALLC all obtain 3 against each other, so selection among them is neutral. They differ in how they respond to the defecting mutants that the mutation operator keeps generating. PAVLOV can be invaded by ALLD: against ALLD it alternates C and D and obtains only 0.5, whereas TFT and GRIM obtain 0.99. The two one-bit mutants of TFT that defect (STFT `10101`, which starts with D, and `01101`, which defects after CC) drag TFT into an alternation of retaliation that gives it a payoff of about 2.5, whereas GRIM punishes forever and obtains only 1.02–1.03. In contrast, against the two one-bit mutants of GRIM that defect (`10111`, which starts with D, and `01111`, which defects after CC), GRIM and TFT obtain exactly the same (0.99 and 1.01). GRIM dominates in the first generations (initial peak in Figure 3a) because the random initial population is full of strategies that do not retaliate, which GRIM exploits (e.g. 4.97 against `10000`, versus 2.99 for TFT). Once the population cooperates, however, TFT displaces GRIM (25 of the 30 runs): against the mutants of TFT it earns much more than GRIM (2.5 versus 1.02–1.03) or PAVLOV (2.0), and against the mutants of GRIM it earns the same.

With moderate noise ($p_{\text{noise}} = 0.01$). Noise breaks the equilibrium of TFT: one error triggers a chain of alternating retaliation, and the payoff of TFT against TFT drops to 2.52. More forgiving variants of TFT, such as ALLC or the one-bit mutants `00001` (which does not retaliate after being exploited) and `00100` (which returns to cooperation after mutual defection), obtain more than TFT against TFT (advantages of +0.43 and +0.20 for the two mutants) and invade it. Such a population is then exploited by GRIM (4.00 against ALLC) and ALLD (4.94). PAVLOV, which obtains much more against itself (2.95) than TFT (2.52) or GRIM



(a) $p_{\text{noise}} = 0$ **(b)** $p_{\text{noise}} = 0.01$ **(c)** $p_{\text{noise}} = 0.05$

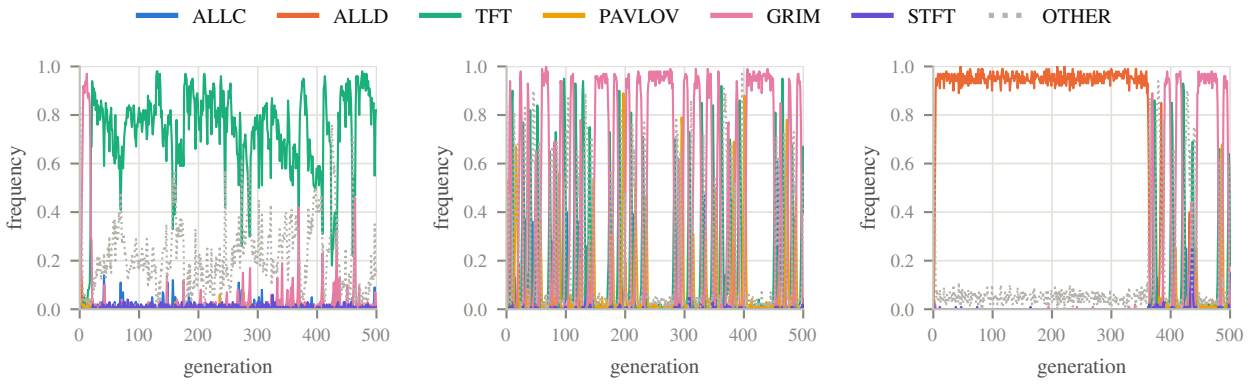
Figure 5: Expected payoff per round of the row strategy against the column strategy (150 rounds). The darker the cell, the higher the payoff.

Table 3: Invasion analysis: named strategies that can invade a population of the resident. "Not invadable" means that none of the 32 memory-1 strategies obtains against the resident a payoff equal to or higher than the resident itself. "None" means that no named strategy invades, but some of the 32 strategies tie with the resident (e.g. ALLC obtains 3 against TFT, like TFT itself), so they can spread by neutral drift.

Resident	$p_{\text{noise}} = 0$	$p_{\text{noise}} = 0.01$	$p_{\text{noise}} = 0.05$
ALLC	ALLD, STFT	all	all
ALLD	none	not invadable	not invadable
TFT	none	ALLC	ALLC
PAVLOV	ALLD	ALLD, GRIM	ALLD, GRIM
GRIM	none	not invadable	not invadable
STFT	ALLC, TFT, PAVLOV, GRIM	ALLC, TFT, PAVLOV	ALLC, TFT, PAVLOV

(1.66), can be invaded by ALLD (2.99) and by GRIM (2.98), which exploit the fact that PAVLOV returns to cooperation after mutual defection. In contrast, GRIM cannot be invaded by *any* of the 32 memory-1 strategies, although TFT is almost tied with it (1.655 versus 1.658). GRIM is therefore the usual attractor (it exceeds 50 % of the population in half of the generations from 100 onwards, pooling all runs), and because it is almost neutral with respect to TFT the population can cycle: GRIM drifts towards TFT, the forgiving variants invade TFT, and GRIM exploits them again. In 22 of the 30 runs there is at least one change of regime, and in the periods dominated by OTHER the most common genomes are precisely `00100` and `00001` (17 % each), followed by TFT and ALLC. The forgiving variants also feed the PAVLOV peaks seen in all 30 runs (its maximum in each run has a median of 82 % and ranges from 59 to 93 %): PAVLOV, one bit away from `00100`, exploits them (3.80 against ALLC, 2.97 against `00100`, which obtains 2.78 against itself) while cooperating with itself (2.95), until ALLD or GRIM invade it. Figure 6b shows these cycles in one run. They produce the cooperation peaks of Figure 1, and they are why the final mean payoff (2.01) is higher than that of a pure GRIM population (1.66).

With high noise ($p_{\text{noise}} = 0.05$). Errors are so frequent (in a pair, one every 10 rounds on average) that no cooperative strategy is stable: reciprocal ones either cooperate poorly with themselves (TFT obtains 2.30 against TFT) or are exploited (PAVLOV by ALLD, 2.93 versus 2.78), and unconditional cooperators such as ALLC (2.95 against itself) are invaded by any defector. Only two non-invadable states remain, ALLD and GRIM, and each is stable against the other: ALLD obtains 1.18 against GRIM, less than GRIM against GRIM (1.31), and GRIM obtains 1.14 against ALLD, less than ALLD against ALLD (1.15). However, the population does not stay fixed in one of them: ALLD exceeds 50 % of the population in 57 % of the generations from 100 onwards (pooling all runs) and GRIM in 34 % (in both cases almost always in stretches of at least 10 generations), and 18 of the 30 runs change regime at least once. Almost neutral mutants make these transitions possible: the one-bit mutants `10111` and `01111` of ALLD lose only 0.002 and 0.006 with respect to ALLD, so they spread by drift, and `10111` (GRIM that starts by defecting, the most frequent genome in the OTHER phases) is a single bit away from GRIM. The TFT peaks (above 30 % in 22 runs) arise inside GRIM phases: TFT is one bit away from GRIM and almost neutral against it (1.301 versus 1.310), so it spreads by drift, and once common it earns more against itself (2.30) than GRIM does against it (1.50); it then drifts towards PAVLOV and forgiving variants, which ALLD and GRIM exploit again. The bimodality



(a) $p_{\text{noise}} = 0$ **(b)** $p_{\text{noise}} = 0.01$ **(c)** $p_{\text{noise}} = 0.05$

Figure 6: Dynamics of a single run per noise level (run 0). With $p_{\text{noise}} = 0.01$ there are cycles between GRIM and cooperative variants; with $p_{\text{noise}} = 0.05$, a long ALLD phase followed by a switch to GRIM.

of Figure 4c is therefore a snapshot of a process that alternates between two defecting strategies (Figure 6c).

Why does PAVLOV not win? In the classical result in favour of PAVLOV [6], obtained with stochastic strategies, PAVLOV only takes over after TFT-like strategies have eliminated the defectors. With deterministic memory-1 strategies, PAVLOV has a structural weakness: against an unconditional defector it cooperates every other round. With our mutation operator, ALLD and GRIM reappear continuously and exploit that weakness before PAVLOV can establish itself.

5 Extension: memory-2 strategies

The exercise asks whether more complex strategies change the conclusion. We repeat the experiment with memory-2 strategies, which decide on the basis of the last two outcomes, and keep everything else the same: the parameters of Table 1, 30 runs per noise level, the same seeds and the same response variable. Runs of the memory 1 and memory 2 with the same seed are unrelated (the genomes differ), so we compare the 30 memory-1 and 30 memory-2 values with the Mann–Whitney test for independent samples.

Representation and payoff computation. A memory-2 strategy has 18 bits: the actions in rounds 1 and 2 (unconditional, as the exercise proposes) and the response to each of the 16 pairs (previous outcome, current outcome). With $2^{18} = 262\,144$ strategies it is no longer possible to name them all or to precompute the full payoff matrix, which would take about 550 GB. We compute the expected payoff with the same exact method, but the state of the Markov chain is now the pair of the last two outcomes (16 states), and we only compute it for the pairs of genomes that appear in the population, caching them during the run. We checked both exact computations against an independent simulator (`report/check_exact.py`): without noise they match exactly (all 1024 memory-1 pairs and 300 random memory-2 pairs), and with noise they agree with the mean of 20 000 simulated matches within Monte Carlo error (all differences below 2 standard errors). We write a genome as `ab | CC CD DC DD`: the first two moves and, for each previous outcome (CC, CD, DC, DD), a group of 4 bits with the response to each current outcome. The named memory-1 strategies are still included: a memory-2 genome is equivalent to one of them if it has the same first move and its four groups equal the last 4 bits of the memory-1 genome, so it ignores the outcome two rounds back. For example, TFT (`00101`) is `0x | 0101 0101 0101 0101`; the second move x is fixed, so it can differ from TFT only in round 2.

Table 4: Mean payoff over the last 100 generations with memory 1 and memory 2 (median of 30 runs, with a 95 % bootstrap CI for memory 2). $\hat{A}_{12}$ is the probability that a memory-2 run beats a memory-1 run; in all three cases, Mann–Whitney $p < 4 \cdot 10^{-11}$.

p_{noise}	Memory 1	Memory 2 [95 % CI]	Difference	$\hat{A}_{12}$
0	2.978	2.913 [2.900, 2.923]	−0.065	0.00
0.01	2.013	2.842 [2.826, 2.844]	+0.829	1.00
0.05	1.171	2.553 [2.551, 2.556]	+1.382	1.00

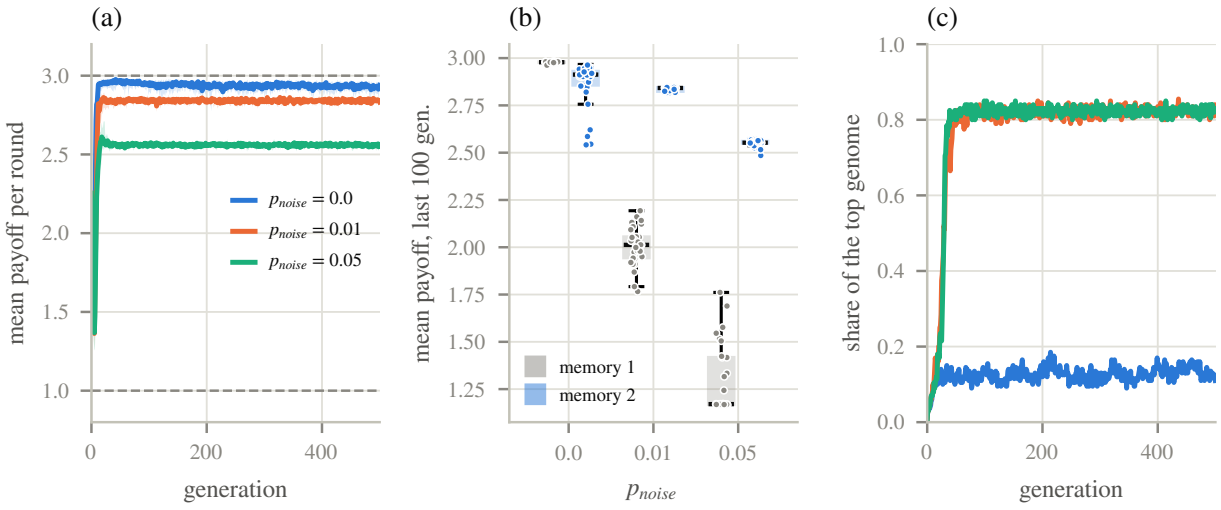


Figure 7: Memory 2. (a) Mean payoff of the population per generation: median and interquartile range of the 30 runs. (b) Response variable of each run with memory 1 (grey) and memory 2 (blue). (c) Share of the population held by the most frequent genome of each run (median over the runs; this genome can differ between runs and over time).

Results. With noise, memory 2 sustains cooperation (Table 4 and Figure 7): with $p_{\text{noise}} = 0.05$ the median payoff is 2.55, against 1.17 with memory 1, and every memory-2 run beats every memory-1 run ($\hat{A}_{12} = 1$). Without noise the difference goes the other way and is small: memory 2 stays slightly lower (2.91 against 2.98) and varies more, with 4 of the 30 runs between 2.54 and 2.62. The population also behaves differently depending on the noise (Figure 7c: for each run and generation we take the share of the most frequent genome, which can differ between runs, and plot the median of the 30 shares). With noise it converges in about 40 generations to a genome that holds 83 % of the population (a median of about 13 distinct genomes), whereas without noise the most frequent genome holds only 15 % and about 43 genomes coexist. Since each particular genome is very rare, the named strategies of Section 3 almost never appear, so we analyse the dominant genomes directly.

Mechanism. *Without noise*, the situations that follow a defection almost never occur among cooperators, so most of the 16 response bits are not under selection and change by drift. Against the most frequent final genome, 21 567 genomes obtain a payoff equal to the one it obtains against itself, or higher. The population does not converge, and drift sometimes makes the dominant genomes exploitable by a particular mutant. In the four collapses we reproduced (`report/check_collapses.py`), the mutant spreads and the population falls into mutual defection (payoff close to 1) within about 7 generations; cooperation returns later (in one run not before generation 500), which explains the runs with a low payoff. Memory 1 has only 5 bits, so this effect is much smaller there.

With noise, errors make every situation occur, and all the bits come under selection. With $p_{\text{noise}} = 0.01$ the most frequent final genome is `10 | 0110 1111 1111 0110` (7 runs; the next two, in 6 and 5 runs, differ from it in one or three bits and cannot be invaded either). If the previous outcome was an agreement (CC or DD), it responds like PAVLOV (`0110`: it cooperates after CC and DD and defects after CD and DC); if it was a disagreement (CD or DC), it always defects. This has two consequences:

- Between copies of itself, an error is followed by two rounds of DD before cooperation resumes (PAVLOV needs only one). It therefore obtains 2.90 against itself, close to PAVLOV (2.95) and well above TFT (2.52) and GRIM (1.66) with memory 1.
- Against ALLD it cooperates only one round in three (PAVLOV, one in two), so ALLD obtains 2.33 against it, less than 2.90, so ALLD cannot invade it. In memory 1, against PAVLOV, ALLD obtains 2.99, more than 2.95 of PAVLOV against itself, which is why PAVLOV could be invaded with memory 1 (Section 4). With memory 2 the strategy can remember that it was exploited and punish one more round while still forgiving, which prevents it from being invaded.

When we play all 262 144 genomes against it, none obtains a payoff equal to or higher than the one it obtains against itself. It is the only best response to itself, so no mutant can invade a population made of it. The same holds with $p_{\text{noise}} = 0.05$ for `11 | 0111 1111 1111 0110` (10 runs, 2.60 against itself). The first two moves barely affect the payoff (each one weighs 1/150), so they are almost neutral.

Not every final genome is like this. In 6 of the 30 runs with $p_{\text{noise}} = 0.05$ the most frequent genome is `11 | 0111 0111 0111 0000`, which cooperates only after CC, like GRIM over one round, except when the previous outcome was DD, after which it always cooperates. This one *can* be invaded: 15 872 genomes obtain against it at least what it obtains against itself (2.65), up to 2.89. In practice it coexists with its one-bit mutant `11 | 0111 0111 0111 0001`, which obtains 0.13 more against it but only 1.97 against itself, so neither takes over. These mixed populations (45–67 % of the most frequent genome at the end) give the lowest memory-2 payoffs with $p_{\text{noise}} = 0.05$ (2.49–2.54).

Conclusion of the extension. With memory 2 the hypothesis of the exercise fits better than with memory 1: with moderate noise PAVLOV-like strategies dominate (H2), although they are harsher than PAVLOV, and with $p_{\text{noise}} = 0.05$ cooperation does not collapse (H3 is refuted). The collapse of cooperation under noise that we observe with memory 1 therefore depends on the strategy space.

6 Limitations

- We only consider deterministic strategies. Section 5 shows that moving from memory 1 to memory 2 changes the conclusion under noise; we have not studied finite state machines or stochastic strategies (*generous TFT*). In memory 2 the second move is unconditional (as the exercise proposes), and the invasion check considers a single mutant in a large population, so it does not rule out that several neutral mutants accumulated by drift destabilise it.
- We use a single set of GA parameters. The population of 100, $p_m = 0.01$ and the tournament of 3 set the strength of drift and selection. Since regime changes depend on almost neutral mutants, a different population size or mutation rate would change their frequency, and perhaps the stability of PAVLOV, which peaks in every run with $p_{\text{noise}} = 0.01$ (up to 59–93 %) but is invaded by very small margins (GRIM 2.98 and ALLD 2.99 against its 2.95).
- Memory 2 without noise is only reproducible on the same machine and software. Without noise many individuals have exactly the same fitness, and differences of 10^{-15} in floating-point arithmetic (which depend on the processor and the NumPy version) decide who wins a tournament. On another computer the same seeds gave a run that diverged at generation 163 and slightly different figures for $p_{\text{noise}} = 0$: median 2.913 (unchanged), 95 % CI [2.896, 2.933], minimum 2.467, about 47 distinct genomes, and a most frequent final genome that is tied, not beaten, by 8 192 genomes (21 568 in our run, some of which exceed it: 3.00 against 2.97). The conclusions do not change ($\hat{A}_{12} = 0$ against memory 1 in both cases). Memory 1 and memory 2 with noise reproduced exactly (up to 10^{-15}).
- An individual does not play against itself. With a population of 100 the effect is small, but not zero.
- The mean payoff mixes cooperation and noise: with high noise, even ALLD against ALLD obtains more than 1, because errors produce involuntary cooperation. This is why we complement the payoff with the composition of the population.
- The definitions of dominance and regime (50 % of the population, 10 generations) are our own conventions; the payoffs and the tests do not depend on them.
- In an additional robustness check (10 runs per configuration, `report/robustness.py`) the conclusions hold with matches of 1000 rounds (TFT dominates in 10/10 without noise, GRIM in 6/10 with $p_{\text{noise}} = 0.01$ and ALLD in 6/10 and GRIM in 2/10 with $p_{\text{noise}} = 0.05$) and with more noise ($p_{\text{noise}} = 0.1$: ALLD in 8/10; $p_{\text{noise}} = 0.25$: ALLD in 10/10, with a payoff of 1.70 that only reflects the noise). We have not explored the levels between 0 and 0.01 systematically, and the change from TFT to GRIM happens somewhere in that range. We have not tested more noise: with $p_{\text{noise}} = 0.5$ every action is random, the four outcomes are equally likely whatever the strategies, and everyone obtains $(3+0+5+1)/4 = 2.25$; and $p_{\text{noise}} > 0.5$ is equivalent to noise $1 - p_{\text{noise}}$ applied to the opposite strategies (all bits flipped).

7 Conclusions

In a population of memory-1 strategies evolving with a GA, cooperation emerges and is stable without noise, with TFT as the dominant strategy. Just 1 % of errors is enough to lose a third of the mean payoff: the population becomes dominated by GRIM and goes through cycles of cooperation and exploitation. With 5 %, cooperation collapses and the population alternates between two defecting states, ALLD and GRIM. The hypothesis holds at the extremes, but the forgiving strategy PAVLOV does not win with moderate noise,

because it can be exploited by ALLD and GRIM, which cannot be invaded by any memory-1 strategy. The invasion analysis on the matrix of expected payoffs, together with the almost neutral mutants, accounts for the cycles and for the bimodality. With memory-2 strategies, on the other hand, cooperation survives the noise: the most frequent final strategies are PAVLOV variants that punish one more round after being exploited, and no mutant can invade the most frequent one. Whether cooperation collapses under noise therefore depends on how much memory the strategies have.

Reproducibility. All the code, seeds and instructions are in `README.txt`: `python main.py` runs the experiment and `python analysis.py` generates all the tables (`memory1/results/summary.txt`) and the figures of this report. For memory 2, `python main_memory2.py` and `python analysis_memory2.py` (results in `memory2/results/summary.txt`).

Use of generative AI tools. The first implementation of the main GA code was written from scratch, and the first interpretation of the results was done without AI. After that, we used Claude (Anthropic, Claude Opus 5.5 model, November 2026) as an assistant: it proposed optimisations and provided scripts to extract the CSV files and plots from the main results, which helped us deepen the experiments and the conclusions.

The main contributions of the AI were the following:

- Use of the exact expected payoff instead of playing a single noisy match per pair, thus ensuring a far more accurate result that does not depend on luck. Done for both memory 1 and memory 2.
- Improvement of the efficiency of memory 2 with a partial cache, since a single run without storing any strategy-pair match took minutes.
- Detection of a reproducibility issue in memory 2.
- Extraction of tables and figures, as well as the proposal of the statistical tests best suited to our results.
- Help with the literature review, for example by proposing [2].
- First draft of the report from the conclusions we extracted, based on the code, tables and plots.
- Correction of grammatical and syntactical errors in the draft.
- Finally, created an interactive web page based on our code and results.

However, we did not take a passive role in these improvements. All of these tasks went through tens of iterations and were constantly reviewed and corrected by us.

This, lets us state that the results were obtained by running our code, and we have reviewed, verified and edited all the code and the final text.

Finally, we want to state that, as MAI students, our focus is to produce the best possible results, and thus, learn from them. We aim for correctness and efficiency, collaborating in the process with powerful AI tools, learning how to use them and always taking an active role in the process.

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